Customer Churn Prediction & Business Value Report

1 Introduction & Project Overview

Customer churn is a critical metric for telecommunication companies. Retaining existing customers is significantly more cost-effective than acquiring new ones. This report documents an end-to-end machine learning project designed to predict customer churn, uncover key behavioral patterns, and translate data insights into actionable business strategies.

2 Data Preprocessing & Feature Engineering

To prevent data leakage and ensure robust model training, the following steps were meticulously implemented:

- **Initial Cleaning:** Dropped irrelevant identifiers (e.g., 'customerID') and cleaned the 'TotalCharges' column, converting it to numeric and filling missing entries.
- **Binary Encoding:** Converted binary categorical features and the target variable ('Churn') directly into numeric values (0 and 1).
- **Train/Test Split:** Performed a stratified split (80% training, 20% testing) with a fixed random state to preserve the target class distribution.
- **Feature Engineering (Post-Split):**
  - *Total_Services:* Computed the cumulative count of active products/services per customer (e.g., streaming, tech support, online security) to capture product stickiness.
  - *Tenure_Group:* Categorized customer tenure into psychological loyalty segments: *New* (≤ 12 months), *Mid-term* ($13 - 36$ months), and *Loyal* (> 36 months).

3 Pipeline Construction & Model Comparison

A unified scikit-learn `Pipeline` and `ColumnTransformer` was built to handle numerical scaling (`StandardScaler`) and remaining multiclass categorical encoding (`OneHotEncoder`) exclusively on training data.

We evaluated five distinct algorithms:

1. **Logistic Regression (Baseline):** F_1-score = 0.5706
2. **Decision Tree:** F_1-score = 0.4946
3. **Random Forest:** F_1-score = 0.5186
4. **Gradient Boosting (Winner):** F_1-score = 0.5727 (Accuracy = 80%)
5. **XGBoost:** F_1-score = 0.5606

4 Task 6: Results, Confusion Matrix, and Business Value

4.1 Confusion Matrix Breakdown

Based on the final optimized **Gradient Boosting** model tested on $1,409$ customer samples:

- **True Negatives (TN = 938):** Customers who truly stayed, correctly identified by the model as non-churners.
- **False Positives (FP = 97):** Loyal customers incorrectly flagged as potential churners (false alarms).
- **False Negatives (FN = 185):** Customers who actually churned, but the model failed to detect them (missed churners).
- **True Positives (TP = 189):** Churning customers correctly caught by the model, enabling proactive intervention.

4.2 Key Behavioral Patterns

- *New Customers at Risk:* The *New* tenure group (≤ 12 months) exhibits the highest rate of abandonment, highlighting the critical nature of the initial onboarding phase.
- *Contract Type Dependency:* Month-to-month contracts strongly correlate with higher churn rates compared to long-term commitments. *Note: Correlation does not prove direct causation, as contract choice reflects broader customer intent and pricing sensitivity.*

4.3 Actionable Business Recommendations

1. **New Customer Onboarding Campaign:** Establish dedicated customer success check-ins, tutorials, and introductory discounts during the first 3 to 6 months to secure early loyalty.
2. **Long-Term Commitment Incentives:** Offer attractive pricing tiers or bundle upgrades to encourage month-to-month subscribers to transition into annual contracts.

4.4 Model Limitations

- **Class Imbalance:** The dataset contains significantly more active users than churners, causing the model to lean slightly toward predicting retention.
- **False Alarms (FP):** Allocating retention budgets to the 97 false-positive customers can incur unnecessary promotional costs.

4.5 Final Recommendation

We formally recommend deploying the **Gradient Boosting** model. It delivers the optimal balance between overall accuracy (80%) and the balanced F_1-score required to detect customer churn effectively without overwhelming operational teams with false alarms.